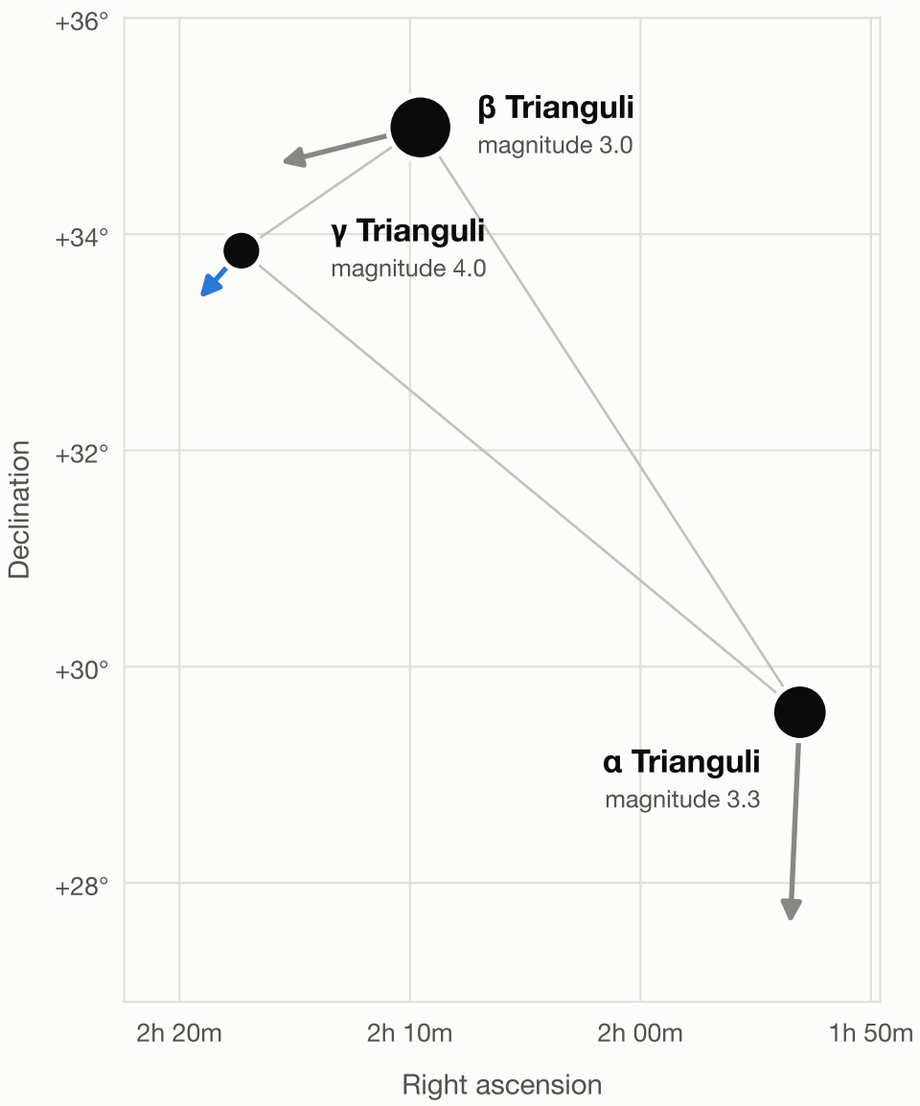
α, β and γ Trianguli in Gaia's latest data release

Positions, distances and motions of the three stars that form the Triangle, from ESA's Gaia Data Release 3 (DR3)

Measured by Gaia DR3 · Not in Gaia DR3, so the Hipparcos value is shown (observed 1989–93, re-analysed 2007) · — *Not available*

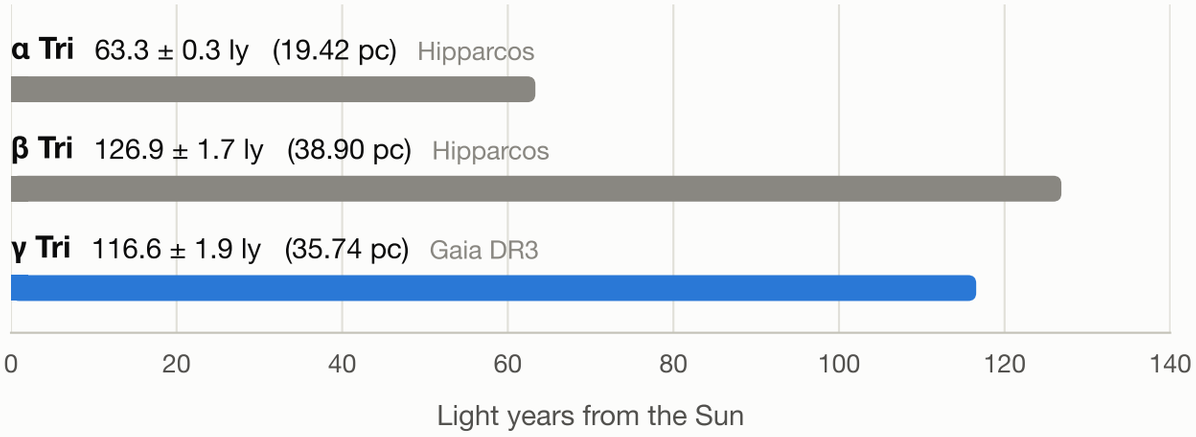
Where they are on the sky

Gaia positions at epoch 2016.0. East is to the left, as on a star chart; dot size shows brightness. Arrows show how far each star drifts across the sky in 30,000 years at its current speed.



How far away they are

Distance = 1 / parallax. The ± uncertainties are too small to show at this scale.



What's in the data, and what's missing

- γ Tri has a complete Gaia record: position, parallax, proper motion and radial velocity (+5.1 km/s, moving away from us).
- α and β Tri are among the brightest stars Gaia observed, and DR3 gives only their positions. Their distances and motions here come from Hipparcos.
- Neither α nor β Tri has a Gaia radial velocity. Both are binary stars, so that speed changes as they orbit.
- γ Tri's Gaia fit is flagged as slightly uncertain (RUWE 1.67; the usual cut-off is 1.4). Hipparcos puts it a little closer, at 112 ly.

The numbers

Cell colour shows where each value comes from.

	α Trianguli Gaia G magnitude 3.31	β Trianguli Gaia G magnitude 3.01	γ Trianguli Gaia G magnitude 4.02
Gaia DR3 source ID	299464296907113984 HIP 8796	329384516560293248 HIP 10064	326109689896765568 HIP 10670
Right ascension ICRS, epoch 2016.0	01h 53m 04.920s 28.270502° ± 0.20 mas	02h 09m 32.821s 32.386754° ± 0.46 mas	02h 17m 18.925s 34.328856° ± 0.35 mas
Declination ICRS, epoch 2016.0	+29° 34′ 40.02″ +29.577783° ± 0.16 mas	+34° 59′ 13.63″ +34.987120° ± 0.41 mas	+33° 50′ 49.07″ +33.846963° ± 0.31 mas
Parallax	51.50 ± 0.23 mas Hipparcos value	25.71 ± 0.34 mas Hipparcos value	27.98 ± 0.46 mas
Distance 1 / parallax	63.3 ± 0.3 light years 19.42 ± 0.09 parsecs · Hipparcos	126.9 ± 1.7 light years 38.90 ± 0.51 parsecs · Hipparcos	116.6 ± 1.9 light years 35.74 ± 0.59 parsecs
Proper motion, RA μα cos δ	+10.82 ± 0.21 mas/yr Hipparcos value	+149.16 ± 0.35 mas/yr Hipparcos value	+45.25 ± 0.33 mas/yr
Proper motion, Dec μδ	-234.24 ± 0.16 mas/yr Hipparcos value	-39.10 ± 0.35 mas/yr Hipparcos value	-52.69 ± 0.36 mas/yr
Radial velocity along our line of sight	*Not in Gaia DR3* binary star, so it varies	*Not in Gaia DR3* binary star, so it varies	+5.1 ± 1.7 km/s moving away from us

Sources: ESA Gaia DR3 (gaiadr3.gaia_source, Gaia archive, Nov 2026); Hipparcos new reduction (van Leeuwen 2007). 1 parsec = 3.26 light years. mas = milliarcsecond (1/1000 of an arcsecond).